Name: _____

WORKSHEET

Finding the gradient, distance and midpoint

1 Find the gradient of the line passing through the following pairs of points.

a (1, 8) and (3, 4)

b (−5, −2) and (7, 1)

c (−1, −25) and (−2, 38)

d (8, −3) and (−12, 12)

e (16, 1) and (41, 1.5)

f $(2t, t)$ and $(7t, -3t)$

2 Find the angle of inclination of a line with the following gradient:

a $m = 1$ _____

b $m = 0.176$ _____

c $m = -4$ _____

d $m = -90$ _____

e $m = \frac{1}{\sqrt{3}}$ _____

f $m = \sqrt{3}$ _____

3 Find the gradient of the line with an angle of inclination of:

a 50.2°

_____

b 26°34′

_____

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c $101°18.6'$

d $63°26.1'$

4 Find the distance between the following points:

a $(-5, 3)$ and $(-1, 0)$

b $(-4, 2)$ and $(8, -3)$

c $(1, 15)$ and $(8, -9)$

d $(-8, -8)$ and $(-8, 1)$

e $(7, -6)$ and $(1, -1)$

f $(-2, -5)$ and $(4, 1)$

5 Find the midpoint of the following points.

a $(2, -6)$ and $(4, -1)$

b $(1, -2.5)$ and $(-3, 7)$

c $(-2, -3)$ and $(0, 4)$

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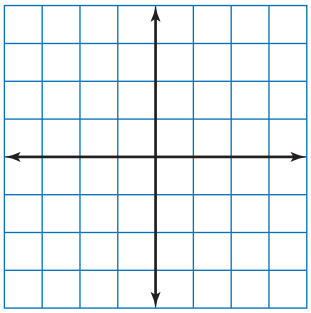
d $(3, 6)$ and $(-1.4, 3.8)$

e $(2t - 3, t + 2)$ and $(5 - 2t, -4 + 7t)$

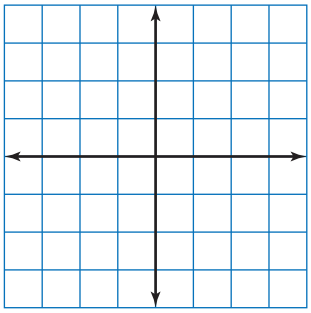
f $(3a, -5b)$ and $(-7a, -b)$

6 If the distance between $(-2, 5)$ and $(s, -1)$ is 10 units, find s.

7 If $(-1, 3)$ is the midpoint of $(x, -4)$ and $(7, y)$, find x and y.



8 Plot the points $(-3, 2)$ and $(3, -1)$ and draw a straight line through them.



a What is the gradient of the line?

b What is the angle of inclination of the line?

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c What is the distance between the two points?

d What is the midpoint of the two points?

e What is the equation of the line?

9 A line passing through two points has a gradient of -0.75 and the distance between the points is 10 units. Give the coordinates of two such points satisfying the following:

a The points are in the 1st and 2nd quadrants.

b Both points are in the 4th quadrant

Answers

1 **a** -2
b 0.25
c -63
d -0.75
e 0.02
f -0.8

2 **a** $45°$
b $10.0°$
c $104.0°$
d $90.6366°$
e $30°$
f $60°$

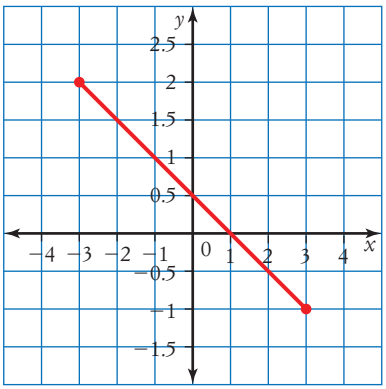
3 **a** 1.2
b 0.5
c -5
d 2

4 **a** 5 units
b 13 units
c 25 units
d 9 units
e 7.8 units
f 8.5 units

5 **a** $\left(3, 2\frac{1}{2}\right)$
b $\left(-1, 2\frac{1}{4}\right)$
c $\left(-1, \frac{1}{2}\right)$
d $(0.8, 4.9)$
e $(1, 4t - 2)$
f $(2a, -3b)$

6 $s = 6, -10$
7 $x = -9, y = 10$

8



a $-\frac{1}{2}$
b $153°26'$
c 6.7 units
d $\left(0, \frac{1}{2}\right)$
e $y = -\frac{1}{2}x + \frac{1}{2}$

9 **a** $(-4, 7)$ and $(4, 1)$ for example
b $(1, -1)$ and $(9, -7)$ for example

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